

BRAINWARE UNIVERSITY

School of Engineering | Dept. of CSE - CS & DS
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A PROJECT REPORT

Submitted in Partial Fulfilment for the Award of the Degree of

Bachelor of Technology

Computer Science & Engineering - Data Science

PROJECT TITLE

Twitter(X) Engagement Patterns: Time, Content & Caption Analysis

Course Code: PROJ-CSD882 | Course Title: Project II

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JUNE 2026

BRAINWARE UNIVERSITY**School of Engineering**

Department of Computer Science & Engineering - Cyber Security & Data Science

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*in partial fulfilment for the award of degree of***Bachelor of Technology in Computer Science & Engineering - Data Science****JUNE 2026**

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled:

"Twitter(X) Engagement Patterns: Time, Content & Caption Analysis"

is the bonafide work of the following students of the Bachelor of Technology in Computer Science & Engineering (Data Science) programme, carried out during the academic year 2025-2026, under my supervision and guidance. The project was submitted to **the Department of Computer Science & Engineering - Cyber Security & Data Science**, School of Engineering, **Brainware University**, in partial fulfilment of the requirements for the award of the degree.

The project report has been checked for plagiarism and the students have carried out the work satisfactorily. I hereby recommend this report for acceptance.

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BRAINWARE UNIVERSITY

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DECLARATION

We hereby declare that the project entitled "Twitter(X) Engagement Patterns: Time, Content & Caption Analysis" submitted to Brainware University in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering Data Science is a record of original work done by us.

The project work has been carried out under the supervision and guidance of our project guide in the Department of Computer Science & Engineering - Cyber Security & Data Science, Brainware University. This project has not been submitted elsewhere for the award of any degree or diploma.

All information and content presented in this report have been collected, processed, and documented in accordance with academic rules and ethical conduct. All external sources, references, libraries, and datasets used in this work have been duly cited and acknowledged.

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ABSTRACT

Keywords: Twitter(X) Analytics · Social Media Engagement · Sentiment Analysis · NLP · TF-IDF · Cosine Similarity · Logistic Regression · Hashtag Analysis · Trend Detection · Content Recommendation

The exponential growth of social media platforms, particularly X (formerly Twitter), has fundamentally transformed the way individuals, organisations, and businesses communicate, share information, and engage with audiences globally. With more than 400 million active users and approximately 500 million tweets generated daily, Twitter has evolved into not just a social networking site but a real-time digital commons -- a space where public opinion forms, trends emerge, and information propagates at unprecedented speed.

Understanding the factors that drive user engagement on such platforms has become increasingly critical for designing effective communication strategies and maximising content visibility. Engagement, defined as the composite measure of likes, retweets, replies, and quote-tweets that a piece of content receives, is influenced by a complex interplay of temporal, linguistic, and contextual variables. Despite the volume of research devoted to individual aspects such as sentiment analysis or hashtag effectiveness, a holistic analytical framework that simultaneously examines posting time, textual content, caption structure, and sentiment dynamics remains scarce.

This project, titled Twitter(X) Engagement Patterns: Time, Content & Caption Analysis, addresses this gap by proposing a comprehensive, data-driven analytical framework designed to study user engagement by examining three critical dimensions: posting time, textual content, and caption structure. The system is built upon a real-world dataset comprising 54,091 tweet records sourced from Kaggle, encompassing 22 attributes including tweet text, datetime stamps, hashtags, likes, retweets, replies, and sentiment labels.

The proposed system employs Natural Language Processing (NLP), statistical analysis, and Machine Learning (ML) techniques to derive meaningful insights. Sentiment analysis is performed using both TextBlob and VADER, enabling classification of tweets into positive, negative, and neutral sentiment categories. Text preprocessing techniques -- including tokenisation, stop-word removal, stemming, and frequency distribution analysis -- are applied to identify trending topics, frequently used keywords, and dominant hashtag patterns.

To understand temporal engagement behaviour, tweet posting activity is analysed across different hours of the day and days of the week to determine optimal posting windows for

maximum audience interaction. A content recommendation module is developed using TF-IDF (Term Frequency-Inverse Document Frequency) and Cosine Similarity, enabling the system to suggest contextually relevant and engaging tweet content based on semantic similarity.

For predictive analysis, a Logistic Regression classification model is implemented for sentiment prediction and evaluated using standard performance metrics including Accuracy (81.86%), Precision, Recall, and F1-Score. Experimental results indicate that integrating sentiment insights, hashtag trends, temporal analysis, and recommendation mechanisms significantly improves understanding of engagement dynamics on Twitter(X).

This project presents a comprehensive analytical framework that can support content creators, digital marketers, researchers, and business organisations in making data-driven decisions to enhance their social media communication strategies, optimise audience engagement, and improve overall online presence.

1. INTRODUCTION

1.1 Background and Motivation

Social media platforms have irrevocably altered the global communication landscape over the past two decades. Among these platforms, Twitter -- now rebranded as X -- occupies a unique and influential position. Unlike platforms that emphasise social connections or multimedia content, Twitter's defining characteristic is its real-time, text-centric format. This design philosophy has made it the platform of choice for breaking news, political discourse, brand engagement, academic discussion, and public opinion formation.

Twitter's architectural simplicity -- short posts of up to 280 characters, amplified through retweet chains, and organised by hashtags -- belies the complex social dynamics that underlie user engagement. A tweet's reach and impact depend not only on the quality of its content but on a constellation of factors including when it is posted, which hashtags it employs, the emotional tone it conveys, and how it resonates with the existing conversation landscape.

The motivation for this project stems from a clear and practically relevant research gap: while individual components of Twitter engagement (sentiment, hashtags, timing) have been studied in isolation, few frameworks integrate these dimensions into a unified analytical and recommendation system. Content creators -- from individual influencers to multinational corporations -- routinely struggle with questions such as: When should I post? What tone generates more engagement? Which hashtags are currently resonating? What type of content should I create?

This project directly addresses these questions by building a data-driven analytical pipeline that processes real Twitter data, extracts meaningful engagement patterns, and generates actionable recommendations.

1.2 The Twitter(X) Platform

Launched in 2006, Twitter grew from a microblogging platform to one of the world's most influential social media services. In 2023, it was rebranded to X under new ownership. Despite the rebrand, the platform retained its core mechanisms -- tweets, retweets, likes, replies, hashtags, and trending topics -- that collectively define the Twitter experience. Key platform characteristics relevant to this study include:

- Character Limit: Posts are restricted to 280 characters, demanding concise expression.

- Real-Time Nature: Tweets appear chronologically and can spread virally within minutes.
- Hashtag System: Hashtags organise content by topic, enabling discovery and trend formation.
- Engagement Metrics: Likes, retweets, replies, and quote-tweets quantify audience response.
- Public API: Twitter has historically provided data access through its API, facilitating research.

1.3 Problem Context

The sheer volume of Twitter data -- approximately 500 million tweets per day -- presents both an opportunity and a challenge. On one hand, this volume provides a rich signal about public opinion, information consumption patterns, and engagement dynamics. On the other, it makes manual analysis impractical. Automated tools that can process this data at scale and extract actionable insights are therefore of significant practical and academic value.

The problem is compounded by the informal, unstructured nature of tweet text. Tweets frequently contain abbreviations, slang, emojis, URLs, @mentions, hashtags, and multilingual content -- all of which require specialised preprocessing before standard NLP techniques can be applied. Furthermore, the transient nature of trends means that any analytical system must be designed for adaptability and potential real-time operation.

1.4 Scope of the Project

This project focuses on the following analytical and computational tasks:

- Sentiment Analysis: Classifying tweets into positive, negative, and neutral categories.
- Trending Topic Detection: Identifying frequently discussed words and themes.
- Hashtag Analysis: Discovering the most impactful hashtags for specific topics.
- Temporal Analysis: Determining peak engagement hours and days.
- Content Recommendation: Suggesting relevant tweets and optimal posting strategies.
- Machine Learning Classification: Training and evaluating a Logistic Regression sentiment classifier.

1.5 Significance of the Study

The significance of this project extends across multiple domains. For digital marketers and brand managers, the system provides data-driven guidance on content strategy and posting schedules. For researchers in computational social science and NLP, it contributes an integrated framework combining multiple analytical approaches. For educators and students in data science, it demonstrates a practical, end-to-end application of NLP, machine learning, and data visualisation techniques on real-world social media data.

The project also aligns with broader trends in the field -- the growing importance of explainable AI, the democratisation of data analysis tools, and the increasing relevance of social media analytics in both commercial and academic contexts.

1.6 Report Organisation

The remainder of this report is organised as follows: Chapter 2 presents a comprehensive literature review. Chapter 3 defines the problem statement. Chapter 4 outlines the project objectives. Chapter 5 describes the dataset. Chapter 6 covers data preprocessing. Chapter 7 details the exploratory data analysis. Chapter 8 presents the methodology. Chapter 9 describes the implementation. Chapter 10 covers model evaluation. Chapter 11 discusses results. Chapter 12 highlights key features and challenges. Chapter 13 concludes the report. Chapter 14 outlines future scope. Chapter 15 lists tools and technologies. Chapter 16 provides references.

2. LITERATURE REVIEW

The study of Twitter engagement patterns sits at the intersection of Natural Language Processing, social network analysis, and machine learning. A substantial body of literature has addressed individual aspects of this problem. This chapter reviews seminal works and recent advances that directly inform the methodology and design of this project.

2.1 Sentiment Analysis on Twitter

2.1.1 Pak and Paroubek (2010)

Pak and Paroubek introduced one of the earliest systematic approaches for sentiment analysis on Twitter. Their foundational work, published in LREC 2010, established the viability of Twitter as a corpus for opinion mining. The authors proposed an automatic labelling method leveraging emoticons (positive emoticons like :) and negative like :-() to build a large-scale, weakly supervised training corpus. After preprocessing steps including tokenisation, stopword removal, and text normalisation, they demonstrated that machine learning algorithms -- specifically Naive Bayes -- could achieve meaningful classification accuracy. This work established Twitter as a reliable source for opinion mining and created the blueprint for much subsequent research in social media sentiment analysis.

2.1.2 Pang, Lee, and Vaithyanathan (2002)

Although predating Twitter, this seminal paper on sentiment classification using machine learning techniques (Naive Bayes, Maximum Entropy, and SVM) remains the foundational reference for ML-based sentiment analysis. Their finding that ML methods substantially outperform lexicon-based approaches established the paradigm that this project follows. The insight that textual features, when vectorised appropriately (e.g., via bag-of-words or TF-IDF), carry sufficient discriminative signal for sentiment classification directly underpins the Logistic Regression model implemented in this project.

2.2 Twitter as Social and News Media

2.2.1 Kwak, Lee, Park, and Moon (2010)

This widely cited study analysed the structural properties of Twitter by examining 41.7 million user profiles and 1.47 billion social relations. The authors demonstrated that Twitter's follower-followee network has properties distinct from both traditional social networks and news media -- characterised by low reciprocity, short topological distances, and rapid information propagation through retweet chains. Their analysis of trending topics revealed that 85% of trending topics are news-related, reinforcing Twitter's role as a real-time news medium. The study's insights into retweet behaviour and hashtag dynamics directly inform this project's approach to trend detection and engagement analysis.

2.2.2 Tumasjan, Sprenger, Sandner, and Welpe (2010)

This influential study investigated whether Twitter discussions could predict political outcomes. Analysing over 100,000 tweets mentioning German political parties, the authors found that tweet volume and sentiment correlated meaningfully with election results. The study demonstrated that combining quantitative metrics (tweet frequency) with qualitative analysis (sentiment polarity) provides

insights beyond what either approach achieves independently. This integrative philosophy directly shapes this project's framework, which combines temporal analysis, sentiment classification, and content analysis.

2.3 Predictive Applications of Twitter Sentiment

2.3.1 Bollen, Mao, and Zeng (2011)

This groundbreaking study demonstrated that collective public mood measured from Twitter could predict stock market movements. The authors applied the Opinion Finder and GPOMS (Google-Profile of Mood States) tools to millions of tweets, identifying mood dimensions including calmness, happiness, and vitality. They found that the calmness dimension specifically showed a 3-day predictive relationship with Dow Jones Industrial Average changes. While the financial prediction aspect is beyond this project's scope, the study established the validity of using Twitter sentiment as a proxy for broader social signals -- a premise central to this project's sentiment analysis component.

2.4 Information Retrieval and Recommendation Techniques

2.4.1 Ramos (2003) -- TF-IDF

Ramos's comprehensive analysis of TF-IDF (Term Frequency-Inverse Document Frequency) as a technique for determining word relevance remains the standard reference for this fundamental NLP method. TF-IDF assigns higher weight to terms that appear frequently in a specific document but infrequently across the broader corpus -- effectively capturing topically distinctive terms. In the context of this project, TF-IDF serves dual purposes: (1) as the feature extraction method for the Logistic Regression classifier, and (2) as the vectorisation basis for the cosine similarity-based recommendation system.

2.4.2 Salton and McGill (1983) -- Vector Space Model and Cosine Similarity

Salton and McGill's landmark work introduced the vector space model and cosine similarity for information retrieval -- techniques that remain foundational in modern NLP. Their insight that text documents can be represented as vectors in a high-dimensional term space, and that similarity can be measured as the cosine of the angle between document vectors, enables efficient semantic matching. This project's recommendation module directly implements TF-IDF vectorisation followed by cosine similarity computation to identify and suggest semantically related tweets.

2.5 Deep Learning Approaches to Social Media Analysis

More recent research has explored deep learning architectures for Twitter analysis. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models (particularly BERT and its variants) have demonstrated superior performance on sentiment classification and topic modelling tasks compared to traditional ML approaches. The Twitter Hybrid Gated Attention Recurrent Network (HGARN) proposed in recent literature achieves state-of-the-art sentiment detection accuracy by combining attention mechanisms with gated recurrent units.

While this project uses Logistic Regression as its primary classifier -- a deliberate choice for interpretability and computational efficiency -- the literature review confirms that future iterations could benefit substantially from BERT-based fine-tuning, which would address the model's current limitations in handling sarcasm, irony, and context-dependent sentiment.

2.6 Benchmark Datasets and Evaluation Frameworks

The SemEval shared task series (particularly SemEval-2015 Task 10 and SemEval-2017 Task 4) has established standard benchmarks for Twitter sentiment analysis. These competitions have driven significant methodological advances and provide reference performance baselines against which this project's results can be contextualised. The Sentiment140 dataset -- 1.6 million automatically labelled tweets -- has served as a standard training resource in numerous studies.

2.7 Research Gap and Project Contribution

The literature review reveals several common gaps that this project addresses:

- Most studies focus on single analytical dimensions (sentiment OR timing OR hashtags) rather than integrating all three.
- Recommendation systems for tweet content generation and optimal posting time are underexplored.
- The combination of real-world dataset analysis with a deployable recommendation module is rare in academic literature.

This project contributes by building an integrated framework that simultaneously addresses sentiment analysis, trend detection, temporal optimisation, and content recommendation -- providing a more holistic tool than existing approaches.

3. PROBLEM STATEMENT

The rapid proliferation of digital communication has elevated social media platforms -- particularly Twitter(X) -- to pivotal roles in public discourse, brand communication, and information dissemination. Twitter generates over 500 million tweets daily, embedding within this colossal stream an extraordinarily rich signal about public behaviour, sentiment, opinion, and engagement dynamics. Yet extracting actionable intelligence from this stream poses formidable challenges that this project seeks to address.

3.1 Core Challenges

3.1.1 Complexity of Engagement Factors

Twitter engagement -- measured through likes, retweets, replies, and quote-tweets -- is not determined by a single factor but by the complex interaction of multiple variables. The timing of a post relative to audience activity peaks, the emotional tone of the content, the strategic use of trending hashtags, the caption structure, and the relevance of the content to ongoing conversations all contribute to how widely and enthusiastically a tweet is received. Understanding the relative contribution and optimal combination of these factors requires a multi-dimensional analytical approach that current tools do not provide in an integrated manner.

3.1.2 Unstructured and Noisy Text Data

Tweet text is inherently messy. Users routinely employ abbreviations, neologisms, slang, internet-specific expressions (e.g., lol , tbh , smh), emojis, hashtags embedded mid-sentence, @mentions, and URLs. Multilingual tweets compound the difficulty. Direct application of standard NLP pipelines to raw tweet text without domain-specific preprocessing yields poor results. The challenge of cleaning, normalising, and standardising this unstructured text while preserving the semantic information it contains is non-trivial.

3.1.3 Temporal Ambiguity in Optimal Posting

Content creators and social media managers operate with incomplete information about when their target audience is most active. While general guidelines exist (e.g., post during business hours), optimal posting times vary significantly by topic, audience geography, industry, and the type of content being shared. The absence of data-driven, topic-specific posting time recommendations forces practitioners to rely on generalised heuristics that may not align with their specific use case.

3.1.4 Hashtag Discovery and Strategic Usage

Hashtags serve as the primary organisational mechanism on Twitter, connecting individual tweets to broader conversations. Selecting appropriate hashtags is a non-trivial problem: overused generic hashtags provide minimal discovery benefit, while highly specific hashtags may fail to connect with any

substantial audience. Topic-specific hashtag analysis that identifies high-signal, well-calibrated hashtags for particular subjects of interest is therefore a genuine and practically relevant challenge.

3.1.5 Absence of Integrated Analytical Frameworks

Existing tools for Twitter analytics -- both commercial and open-source -- tend to focus on one or two analytical dimensions. Sentiment analysis tools do not typically incorporate temporal analysis. Hashtag tracking platforms rarely integrate content quality assessment. This fragmentation means that users must navigate multiple disconnected tools and manually synthesise insights -- a time-consuming and error-prone process that limits the accessibility of data-driven social media strategy.

3.1.6 Limited Intelligent Recommendation Support

Perhaps most critically, existing analytical frameworks are predominantly descriptive rather than prescriptive. They report what has happened but provide limited guidance on what should be done. A content recommendation system that synthesises insights from sentiment analysis, trend detection, and temporal analysis to suggest specific tweet content, hashtags, and posting times would offer substantially greater practical value than purely descriptive analytics.

3.2 Problem Formalisation

The formal problem addressed by this project can be stated as follows: Given a corpus of historical tweets $C = \{t_1, t_2, \dots, t_n\}$ where each tweet t_i has associated text, timestamp, hashtags, and engagement metrics, develop an integrated analytical system that:

- Accurately classifies tweet sentiment (positive, negative, neutral) using NLP and ML techniques.
- Identifies trending topics and high-impact hashtags from the corpus.
- Determines temporal engagement patterns to identify optimal posting windows.
- Recommends relevant tweet content and posting strategies for user-specified topics.
- Evaluates all components using standard performance metrics.

4. PROJECT OBJECTIVES

The primary objective of this project is to develop an intelligent analytical framework capable of studying engagement patterns on X (formerly Twitter) by examining tweet timing, textual content, and caption characteristics. The system is designed to process large volumes of tweet data efficiently and generate meaningful insights that can support data-driven social media strategies.

4.1 Primary Objectives

1. To collect, organise, and preprocess a large-scale Twitter dataset containing sufficient diversity of topics, sentiment, and temporal distribution to support robust analysis.
2. To implement a complete NLP preprocessing pipeline that effectively handles the unique characteristics of tweet text, including noise removal, normalisation, tokenisation, and stop-word elimination.
3. To perform accurate sentiment analysis classifying tweets into positive, negative, and neutral categories, achieving satisfactory precision, recall, and F1-score across all classes.
4. To conduct comprehensive time-series analysis on tweet activity, identifying statistically meaningful patterns in posting behaviour, peak engagement hours, and temporal engagement distributions.
5. To extract, rank, and contextualise hashtag usage patterns, both globally across the dataset and specifically for user-defined topics of interest.
6. To build a content recommendation system using TF-IDF vectorisation and cosine similarity that retrieves and presents contextually relevant tweet content, suggested hashtags, and optimal posting times.
7. To train, evaluate, and validate a Logistic Regression classification model using standard performance metrics, achieving at least 80% overall accuracy on the held-out test set.
8. To create meaningful visualisations of all analytical outputs, including sentiment distributions, trend analyses, temporal patterns, and model performance metrics.

4.2 Secondary Objectives

- To identify and document the key challenges involved in processing real-world social media data at scale.

- To establish a modular system architecture that facilitates easy extension and modification of individual analytical components.
- To demonstrate the practical applicability of the framework through concrete use-case examples.
- To provide a foundation upon which future enhancements (deep learning models, real-time APIs, multi-platform support) can be built.

4.3 Success Criteria

Objective	Success Metric
Sentiment Classifier Accuracy	≥ 80% overall accuracy on test set
Preprocessing Pipeline	Effective noise removal with semantic content retained
Temporal Analysis	Identification of statistically significant peak hours
Hashtag Analysis	Top-10 hashtag extraction per topic with frequency ranking
Recommendation System	Generation of contextually relevant tweet suggestions
Visualisation	Clear graphical representation of all key analytical outputs

5. DATASET DESCRIPTION

5.1 Overview

The dataset used in this project consists of real-world tweet data collected from Kaggle for analysing engagement patterns, textual sentiment, hashtag usage, and temporal activity trends. It represents a rich, multi-dimensional corpus of social media content that provides the empirical foundation for all analytical components of this project.

Dataset Summary
Source: Kaggle (Twitter/X Public Tweet Data)
Total Records: 54,091 tweets
Total Attributes: 22 columns
Data Types: Structured metadata + Unstructured tweet text
Sentiment Classes: Positive / Negative / Neutral
Preprocessing Required: Yes (significant cleaning and normalisation needed)

5.2 Attribute Description

The dataset contains 22 attributes spanning four broad categories: tweet identity, content, engagement metrics, and metadata.

Attribute	Data Type	Description
Datetime	DateTime	Exact timestamp of tweet publication; used for temporal analysis and hour/day extraction.
Tweet Id	Integer	Unique numeric identifier for each tweet.
Text	String	Full raw text content of the tweet; primary input for NLP and sentiment analysis.
Username	String	Account name that posted the tweet.

ReplyCount	Integer	Number of direct replies received; measures conversational engagement.
RetweetCount	Integer	Number of retweets; measures content virality and reach.
LikeCount	Integer	Number of likes (hearts); measures positive reception.
QuoteCount	Integer	Number of quote-tweets; measures content importance.
Language	String	Language code of the tweet (e.g., 'en', 'es').
Source	String	Platform/device from which the tweet was posted.
MentionedUsers	String	List of @mentioned user accounts in the tweet.
QuotedTweet	String	Reference to any tweet being quoted.
Media	String	Information about attached media (images, videos, links).
hashtag	String	Extracted hashtags from tweet text.
hashtag_counts	Integer	Number of hashtags used in the tweet.
Hour	Integer	Hour of posting (0-23), extracted from Datetime for temporal analysis.
Actual_Sentiment	String	Ground-truth sentiment label (Positive/Negative/Neutral) generated using VADER.

5.3 Dataset Characteristics

5.3.1 Volume and Scale

With 54,091 records, the dataset provides sufficient volume for statistically robust analysis. The number of records allows for an 80:20 train-test split (approximately 43,272 training and 10,819 test records) while maintaining adequate representation across all sentiment classes. The scale also supports meaningful temporal analysis across multiple days, hours, and posting patterns.

5.3.2 Content Diversity

The dataset spans multiple topics, languages, and content types, including news commentary, product discussions, event-driven content, and opinion expression. This diversity ensures that the analytical framework is generalised rather than overfitted to a narrow topical domain. Specific topic analysis (e.g., searching for `ipl` or `microsoft`) reveals thousands of relevant tweets, confirming sufficient topical density.

5.3.3 Temporal Distribution

The datetime attribute spans multiple days, enabling analysis of both intra-day patterns (hour-of-day posting distributions) and inter-day patterns (day-of-week activity trends). The temporal granularity at the hour level allows identification of distinct peak engagement windows.

5.3.4 Engagement Metrics

The combination of likes, retweets, replies, and quote-tweets provides a multi-dimensional view of engagement. Each metric captures a distinct aspect of audience response: likes indicate passive appreciation, retweets indicate active endorsement and distribution, replies indicate conversational engagement, and quote-tweets indicate content significance sufficient to warrant commentary.

5.4 Data Quality Observations

Initial inspection of the raw dataset revealed several quality issues that necessitated preprocessing:

- **Missing Values:** Some records contained null or empty values in the Text, Datetime, and engagement metric columns.
- **Mixed Datetime Formats:** Timestamps appeared in multiple formats with inconsistent timezone representations (UTC and local offsets).
- **Embedded Noise in Text:** Tweet text contained URLs, @mentions, #hashtags embedded mid-text, HTML entities, and emoji characters.
- **Multilingual Content:** A proportion of tweets were in languages other than English, requiring language filtering or handling.
- **Duplicate Records:** Some tweet records appeared multiple times in the raw dataset.

6. DATA PREPROCESSING

Data preprocessing is the critical first stage in the analytical pipeline, transforming raw, noisy Twitter data into a clean, structured format suitable for downstream NLP, machine learning, and statistical analysis. The quality of all subsequent analysis depends directly on the effectiveness of this stage. The preprocessing pipeline implemented in this project consists of eight sequential stages.

6.1 Data Loading and Initial Inspection

The raw dataset is loaded from Excel format (.xlsx) using the Pandas library. Initial inspection involves checking the dimensions of the dataframe (54,091 rows × 22 columns), examining data types for each attribute, identifying columns with missing values, computing basic summary statistics for numeric columns, and reviewing sample records to understand content characteristics.

```
import pandas as pd
# Load dataset
data = pd.read_excel('merged_output.xlsx', engine='openpyxl')
print(f'Shape: {data.shape}')
print(data.dtypes)
print(data.isnull().sum())
print(data.head(5))
```

6.2 Handling Missing Values

Records with missing entries in critical columns are handled as follows: rows with null values in the Text column are removed, as tweet text is the primary analytical input; rows with null Datetime values are dropped, as temporal analysis requires valid timestamps; rows with missing engagement metrics (LikeCount, RetweetCount, ReplyCount) are retained but flagged, as these are not critical for all analytical tasks.

After this stage, the dataset is confirmed to have complete data for all columns required by downstream analytical modules. The number of records removed for data quality reasons is tracked and reported.

6.3 Text Cleaning and Normalisation

Raw tweet text undergoes a systematic cleaning process to remove elements that add noise without contributing semantic value to the analysis. The cleaning steps are applied sequentially using Python's re (regular expressions) module:

- URL Removal: All HTTP and HTTPS URLs are replaced with empty strings using the pattern `r'http\S+'`.
- Mention Removal: @username mentions are removed using the pattern `r'@\w+'`.
- Hashtag Stripping: In some analysis contexts, the '#' symbol is removed from hashtag tokens (while the hashtag text itself is retained for keyword analysis).
- Special Character Removal: Non-alphanumeric characters (excluding spaces) are removed.

- Lowercasing: All text is converted to lowercase to ensure case-insensitive token matching.

Whitespace Normalisation: Multiple consecutive spaces are collapsed to single spaces.

import re

def clean_text(text):

```
text = str(text)  
text = re.sub(r'http\S+', '', text)    # Remove URLs  
text = re.sub(r'@\w+', '', text)      # Remove mentions  
text = re.sub(r'#(\w+)', r'\1', text) # Strip # symbol  
text = re.sub(r'^\w\s', '', text)     # Remove special chars  
text = text.lower().strip()           # Lowercase & strip  
return ' '.join(text.split())         # Normalise whitespace
```

- **data['Clean_Text'] = data['Text'].apply(clean_text)**

6.4 Stopword Removal and Tokenisation

After text cleaning, tweets are tokenised into individual words using a simple whitespace-based split. English stopwords from the NLTK corpus are then eliminated. Stopwords are common function words (e.g., the, is, and, to) that appear with high frequency across all documents and carry minimal discriminative information for topic and sentiment analysis. Their removal reduces feature dimensionality and improves signal-to-noise ratio.

A custom stopwords list is also maintained to handle Twitter-specific tokens (e.g., rt for retweet, amp for the HTML entity &) that NLTK's standard list does not cover.

6.5 Datetime Processing

The Datetime column presents a significant preprocessing challenge due to mixed format timestamps and timezone representations. The processing steps are:

- Format Parsing: Pandas `pd.to_datetime()` with `format='mixed'` and `errors='coerce'` handles the variety of timestamp formats.
- Timezone Handling: `utc=True` normalises all timestamps to UTC.
- Invalid Entry Removal: Records with unparseable timestamps (coerced to NaT) are dropped.
- Timezone Stripping: For downstream analysis convenience, timezone information is removed after normalisation using `dt.tz_convert(None)`.
- Feature Extraction: Hour (0-23) and day-of-week (0-6, Monday=0) features are extracted for temporal analysis.

TIME_COL = 'Datetime'

```
data[TIME_COL] = pd.to_datetime(  
    data[TIME_COL],  
    format='mixed',  
    errors='coerce',  
    utc=True  
)  
data = data.dropna(subset=[TIME_COL])  
data[TIME_COL] = data[TIME_COL].dt.tz_convert(None)  
data['Hour'] = data[TIME_COL].dt.hour  
data['DayOfWeek'] = data[TIME_COL].dt.day_name()
```

6.6 Sentiment Label Generation

To provide ground-truth labels for machine learning model training, sentiment labels are generated programmatically using VADER (Valence Aware Dictionary and sEntiment Reasoner) from NLTK. VADER is specifically calibrated for social media text and handles emoticons, slang, and punctuation patterns that general sentiment tools miss. Tweets are classified as:

- Positive: VADER compound score ≥ 0.05
- Negative: VADER compound score ≤ -0.05
- Neutral: $-0.05 < \text{compound score} < 0.05$

In addition, TextBlob polarity scores are used for real-time sentiment classification in the analytical modules, providing a complementary perspective. The VADER-generated labels form the Actual_Sentiment column used for training the Logistic Regression classifier.

6.7 Feature Transformation for Machine Learning

For the machine learning classification task, tweet text must be converted from string format to numerical feature vectors. TF-IDF (Term Frequency-Inverse Document Frequency) vectorisation is applied using scikit-learn's TfidfVectorizer with English stopwords removal and a maximum feature size calibrated to the corpus. The resulting sparse matrix serves as input to the Logistic Regression classifier.

6.8 Preprocessing Summary

#	Step	Output
1	Data Loading	Pandas DataFrame with 54,091 rows, 22 columns
2	Missing Value Handling	Complete records for Text and Datetime columns
3	Text Cleaning	Noise-free, lowercased tweet text
4	Stopword Removal & Tokenisation	Meaningful keyword tokens per tweet
5	Datetime Processing	Standardised timestamps with Hour and DayOfWeek features
6	Sentiment Label Generation	Actual_Sentiment column (Positive/Negative/Neutral)
7	TF-IDF Vectorisation	Sparse numerical feature matrix for ML training

7. METHODOLOGY

The methodology adopted in this project follows a systematic and data-driven framework for Analysing the engagement patterns on X (formerly Twitter) by examining tweet timing, textual content, sentiment, and hashtag usage. The complete workflow is designed to transform raw tweet data into meaningful analytical insights through preprocessing, feature extraction, exploratory analysis, and machine learning techniques. The methodology consists of the following stages:

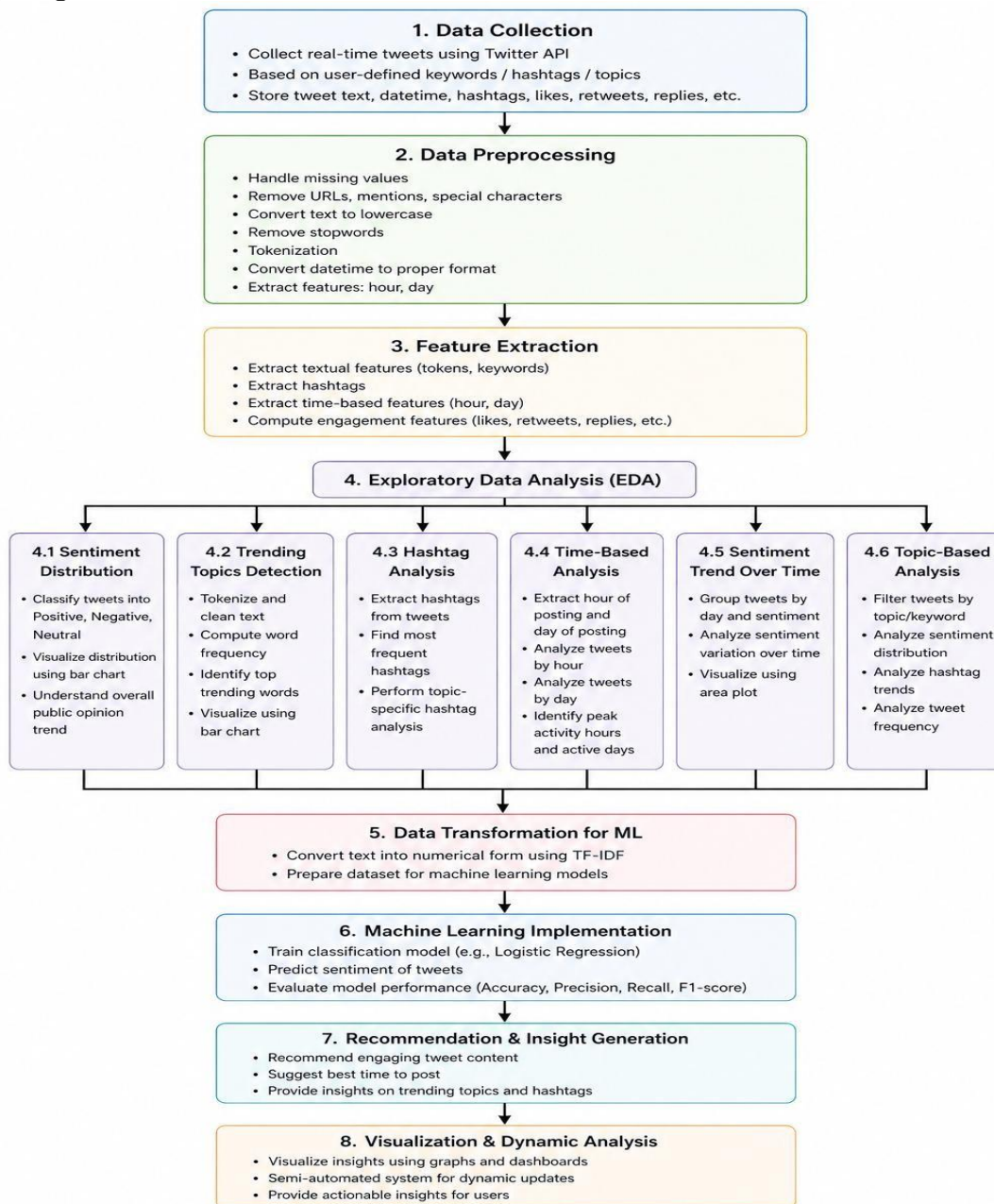


Figure 1: Flowchart Representing the Proposed Methodology

7.1 Data Collection

Data collection is the first and one of the most important stages of the project. In this phase, tweet data is gathered from social media platforms, mainly Twitter/X, using predefined keywords, hashtags, usernames, or trending topics. The purpose of collecting this data is to analyze user engagement, identify trends, and understand public opinion regarding specific topics or events. The dataset contains various types of tweet-related information such as tweet text, date and time of posting, hashtags, likes, retweets, replies, views, and user metadata. These details help in measuring the popularity and engagement level of tweets.

The data collection process is generally performed using APIs, web scraping tools, or publicly available datasets. APIs allow the system to access real-time tweet information directly from the platform. Keywords and hashtags play a major role in filtering relevant tweets from millions of available posts. For example, if the project is focused on sports trends, hashtags related to tournaments, teams, or players are used to gather related tweets. Similarly, for marketing analysis, brand names and campaign hashtags are selected.

Collected data may include structured and unstructured content. Structured data refers to information such as timestamps, retweet counts, and user IDs, while unstructured data mainly consists of tweet text and comments. The collected dataset is stored in databases or CSV files for further processing and analysis. Proper data collection ensures that the dataset is large, diverse, and relevant enough to produce accurate results during analysis and prediction stages.

Another important aspect of data collection is maintaining data quality and relevance. Duplicate tweets, spam content, and irrelevant information are filtered out during or after the collection process. The quality of collected data directly affects the performance of the project because inaccurate or incomplete data can lead to incorrect predictions and analysis. Therefore, careful selection of keywords and proper data extraction methods are necessary for building a reliable and effective system.

7.2 Data Preprocessing

Data preprocessing is the process of cleaning and transforming raw collected data into a structured and usable format. Since social media data often contains noise, incomplete information, and unnecessary text elements, preprocessing is essential for improving data quality before applying machine learning or Natural Language Processing techniques.

The first step in preprocessing is handling missing or null values. Some tweets may not contain hashtags, replies, or complete user information. Such missing data is either removed or replaced with suitable values to maintain dataset consistency. Duplicate entries are also identified and removed to avoid repeated information in the analysis.

After handling missing data, text cleaning is performed. Tweets usually contain URLs, mentions, emojis, punctuation marks, and special characters that do not contribute significantly to text analysis. These

unwanted elements are removed using text processing techniques. For example, links to websites, user mentions starting with “@”, and unnecessary symbols are eliminated to keep only meaningful textual information.

The next preprocessing step involves converting all text into lowercase format. This helps maintain uniformity because words written in uppercase and lowercase are treated differently by machine learning algorithms. For example, “Trend” and “trend” become the same word after conversion, reducing redundancy in the dataset.

Stopword removal is another important task in preprocessing. Stopwords are commonly used words such as “is”, “the”, “and”, and “of” that do not provide significant meaning for analysis. Removing these words helps focus only on important keywords and improves computational efficiency. After stopwords removal, tokenization is applied. Tokenization breaks long text sentences into smaller units called tokens or words. These tokens are easier to analyze and process during Natural Language Processing tasks.

Stemming and lemmatization techniques may also be applied to reduce words to their root forms. For example, words like “running”, “runs”, and “ran” can be converted into the base word “run”. This helps reduce data complexity and improves the accuracy of text analysis models.

Data preprocessing also includes formatting date and time information into standardized formats. Datetime values are converted into machine-readable formats so that temporal analysis can be performed effectively. Clean and properly preprocessed data helps improve model performance, reduces noise, and increases the accuracy of predictions and trend identification.

7.3 Feature Extraction

Feature extraction is the process of identifying and selecting important attributes from the preprocessed dataset that can be used for analysis and prediction. In machine learning and data analytics projects, features are the variables or characteristics that help models understand patterns within the data. Effective feature extraction improves model performance and enhances the accuracy of trend analysis.

One of the primary features extracted from tweet data is textual information. Important keywords, hashtags, phrases, and frequently occurring terms are identified from the tweet text. These textual features help determine the main topics being discussed by users. Hashtags are especially useful because they represent trending subjects and categorize tweets into specific themes. Word frequency analysis and techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) are commonly used to measure the importance of words within the dataset.

Temporal features are another major category of extracted information. Datetime values are analyzed to identify posting patterns based on hour, day, week, or month. These features help determine when users are most active and when engagement levels are highest. For example, tweets posted during peak hours may receive more likes and retweets compared to tweets posted late at night. Temporal analysis is valuable for predicting engagement trends and optimizing posting schedules.

Engagement-related features are also extracted from the dataset. Metrics such as number of likes, retweets, replies, shares, and views are important indicators of tweet popularity. These features help analyze user interaction and measure the success of particular tweets or topics. High engagement values often indicate viral or trending content.

Sentiment-based features can also be generated during feature extraction. Using Natural Language Processing techniques, tweets are classified into positive, negative, or neutral sentiments. Sentiment analysis helps understand public opinion and emotional reactions toward specific topics, products, or events. This information is particularly useful for businesses, marketers, and researchers who want to study audience behavior.

User-related features such as follower count, account verification status, and user activity level may also be included. These attributes help determine the influence and credibility of users posting the tweets. Influential users with large follower bases often generate higher engagement and wider reach.

Feature extraction transforms raw textual and numerical data into meaningful representations that can be easily understood by machine learning algorithms. Properly extracted features improve prediction accuracy, reduce processing complexity, and help generate better insights from the dataset. This stage forms the foundation for trend analysis, sentiment prediction, recommendation systems, and other advanced analytical tasks used in the project.

7.4 EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is a critical investigative phase that precedes formal modelling. It involves applying statistical and visual techniques to understand the dataset's structure, distributions, relationships, and anomalies. In this project, EDA encompasses six complementary analytical perspectives, each addressing a specific dimension of Twitter engagement.

7.4.1 Sentiment Distribution Analysis

The first EDA task examines the overall distribution of sentiment across the dataset. After applying TextBlob sentiment classification to all 54,091 tweets, the distribution was computed and visualised as a bar chart (Figure 1).

Observations from the sentiment distribution reveal that the majority of tweets in the dataset are classified as Neutral (~27,000-28,000 tweets), which is consistent with Twitter's role as an information-sharing platform where many tweets are factual or announcement-based rather than opinion-laden. Positive tweets constitute approximately 15,000 records, while Negative tweets are the smallest category at around 5,000-6,000. This class imbalance has implications for classifier training and evaluation, particularly regarding the F1-score for the negative class.

The preponderance of neutral tweets reflects a broader pattern observed across social media sentiment studies -- the majority of posts are informational or conversational rather than strongly opinionated. This

finding has practical implications: content creators seeking to drive engagement through emotional resonance must work deliberately against this baseline tendency toward neutrality.

7.4.2 Trending Topic Detection

After text cleaning and stopwords removal, word frequency analysis using Python's Counter module identifies the most commonly occurring terms across the corpus. These high-frequency keywords constitute the trending topics of the dataset -- the subjects that dominate the conversation.

Figure 2 presents the top 10 trending words. The term chatgpt (appearing approximately 27,000 times) dominates the corpus, followed by #chatgpt as a hashtag token, ok , ai , chat , and terms related to computing and Microsoft. This word distribution reflects the dataset's topical composition and confirms that AI-related content was a dominant theme during the data collection period -- a finding consistent with the explosive public interest in large language models in 2023.

The trending topic analysis provides a quick snapshot of the discourse landscape, allowing users to contextualise their own content relative to ongoing conversations and identify opportunities for topical alignment.

7.4.3 Hashtag Analysis

Hashtags are the primary organisational mechanism on Twitter. The hashtag analysis module extracts all hashtag tokens from tweets (words beginning with #), computes their frequency distribution, and identifies the most impactful tags both globally and for specific topics.

Figure 3 illustrates the most used hashtags for the topic microsoft . The analysis reveals that #chatgpt (frequency ~630) is the most associated hashtag, followed by #microsoft (~500), #openai (~280), #ai (~240), #artificialintelligence , #technology , #openai , #tech , #google , and #news . This pattern reflects the interconnected discourse around AI and technology topics -- conversations about Microsoft are inseparable from discussions of OpenAI, ChatGPT, and artificial intelligence broadly.

The topic-specific hashtag analysis provides directly actionable guidance: a user creating content about Microsoft should incorporate #chatgpt, #openai, and #ai alongside #microsoft to maximise discoverability within the related conversation network.

7.4.4 Time-Based Engagement Analysis

Temporal analysis examines when Twitter users are most active and when specific topics attract the most discussion. The Hour feature, extracted from the Datetime column, enables grouping of tweets by hour-of-day to construct activity profiles.

Figure 6 (Best Time to Post for microsoft) reveals a strongly bimodal activity pattern. Tweet activity for this topic is relatively low between midnight and approximately 14:00 (2:00 PM), rises sharply around 15:00 (3:00 PM), peaks dramatically at approximately 21:00 (9:00 PM) with close to 500 tweets in a single hour, then declines through the late evening. This pattern likely reflects a US-centric audience posting during the US business afternoon and early evening hours.

The practical implication is clear: content creators targeting technology-focused audiences with Microsoft-related content should schedule posts between 15:00 and 22:00 for maximum visibility. The system's best-time recommendation feature automates this analysis for any user-specified topic.

7.4.5 Sentiment Trend Over Time

Beyond the overall sentiment distribution, understanding how sentiment evolves over time provides insights into dynamic audience behaviour. The sentiment timeline analysis (Figure 5) groups tweets by day and sentiment category, creating a stacked area chart that visualises sentiment evolution.

The sentiment timeline for mobile topics (Figure 5) reveals a period of predominantly positive sentiment with an underlying neutral baseline. The area chart format makes it easy to visually track whether a topic's public reception is trending positive, negative, or shifting -- valuable for crisis communication monitoring and campaign performance tracking.

7.4.6 Topic-Based Focused Analysis

All analytical modules support topic-specific filtering, allowing users to conduct focused analysis on subjects of interest. When a user queries a specific topic (e.g., ipl , microsoft , chatgpt), the system:

- Filters the dataset to tweets containing the topic keyword (case-insensitive).
- Recomputes sentiment distributions for this filtered subset.
- Identifies the most relevant hashtags for this topic context.
- Determines the optimal posting time for this specific topic's audience.
- Generates a sentiment timeline specific to the topic.

This topic-specific analysis ensures that recommendations are calibrated to the actual behaviour of the audience interested in that particular subject, rather than relying on dataset-wide generalisations.

8. Post Methodology Stages:

8.1 Overall System Architecture

The system is designed around eight primary stages:

Stage	Module	Key Activities
1	Data Collection	Kaggle dataset acquisition; tweet text, metadata, timestamps
2	Data Preprocessing	Missing value handling, text cleaning, stopword removal, datetime parsing
3	Feature Extraction	Keywords, hashtags, temporal features, engagement metrics
4	Exploratory Data Analysis	Sentiment distribution, trend detection, hashtag & temporal analysis
5	Data Transformation for ML	TF-IDF vectorisation, train-test split preparation
6	Machine Learning	Logistic Regression training, sentiment prediction, evaluation
7	Recommendation System	Content generation, hashtag suggestion, optimal time recommendation
8	Visualisation	Bar charts, line graphs, area plots, interactive dashboards

8.2 Sentiment Analysis Methodology

Sentiment analysis is implemented using a dual-tool approach that leverages both lexicon-based and ML-based methods:

8.2.1 TextBlob Polarity Analysis

TextBlob computes a polarity score for each tweet in the range [-1.0, +1.0] based on a pattern-matching approach using a pre-built sentiment lexicon. The classification thresholds are: Positive (polarity > 0), Negative (polarity < 0), and Neutral (polarity = 0). TextBlob is used for real-time sentiment scoring in the analytical and recommendation modules due to its computational efficiency.

8.2.2 VADER Sentiment Analysis

VADER (Valence Aware Dictionary and sEntiment Reasoner) is specifically designed for social media text and is substantially more accurate than TextBlob for tweet sentiment due to its handling of capitalisation, punctuation, emoji, and common social media expressions. VADER's compound score is used to generate the Actual_Sentiment labels that serve as ground truth for ML model training. The compound score ranges from -1 (most negative) to +1 (most positive), with thresholds at ± 0.05 for classification.

8.3 TF-IDF and Cosine Similarity

The recommendation system employs a two-step mathematical approach:

Step 1 -- TF-IDF Vectorisation: Each tweet is represented as a vector in a high-dimensional term space. The weight of each term is computed as:

$$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

where $TF(t, d)$ is the frequency of term t in document d , and $IDF(t) = \log(N / df(t))$ measures the inverse document frequency of t across the corpus of N documents. This weighting scheme assigns high scores to terms that are frequent within a document but rare across the corpus -- effectively capturing topically distinctive keywords.

Step 2 -- Cosine Similarity: Given a user-specified topic query q , the system computes the cosine similarity between the query vector and all tweet vectors in the TF-IDF matrix:

$$\cos(q, d) = (q \cdot d) / (\|q\| \times \|d\|)$$

The top- k tweets with highest cosine similarity to the query are returned as recommendations. This approach efficiently identifies semantically related content without requiring exact keyword matches.

8.4 Logistic Regression Classification

Logistic Regression is selected as the primary classification algorithm for sentiment prediction based on three key considerations: (1) its strong baseline performance on high-dimensional sparse text feature matrices, (2) its computational efficiency for training and inference on large datasets, and (3) its inherent interpretability -- the model's coefficient weights provide insight into which terms most strongly predict each sentiment class.

The model is trained on TF-IDF feature vectors derived from tweet text, with the `Actual_Sentiment` column as the target variable. Multi-class classification is handled using a one-vs-rest (OvR) strategy. The maximum number of iterations is set to 200 to ensure convergence on the high-dimensional feature space.

8.5 Recommendation System Design

The tweet recommendation system integrates three analytical components:

- Content Recommendation: TF-IDF + cosine similarity identifies the top-5 most semantically relevant tweets for a given topic and sentiment preference.
- Hashtag Recommendation: Hashtags are extracted from the top recommended tweets and ranked by frequency, providing a data-driven set of high-relevance hashtags.
- Temporal Recommendation: The hour with peak tweet activity for the specified topic and sentiment is identified and presented as the optimal posting time in human-readable AM/PM format.

9. IMPLEMENTATION

The implementation of the project was carried out entirely in Python, leveraging a rich ecosystem of data analysis, NLP, and machine learning libraries. The system was developed in a modular manner within Jupyter Notebook / Google Colab environments, with each analytical module implemented as a self-contained, independently executable script that reads from and writes to shared Excel data files.

9.1 Development Environment

Component	Details
Programming Language	Python 3.x
IDE / Notebook	Jupyter Notebook / Google Colab
Primary Data Format	Microsoft Excel (.xlsx)
Version Control	Git (recommended)
Hardware	Standard CPU (no GPU required for Logistic Regression)
Operating System	Windows (cross-platform compatible)

9.2 Module 1 -- Overall Public Opinion Analysis

This module performs sentiment analysis across the entire dataset and visualises the distribution of positive, negative, and neutral tweets. It serves as the entry-point analysis, providing an immediate overview of the sentiment landscape.

```
import pandas as pd
from textblob import TextBlob
import matplotlib.pyplot as plt

data = pd.read_excel('merged_output.xlsx')
tweets = data['Text']
sentiments = []

for tweet in tweets:
    analysis = TextBlob(str(tweet))
    if analysis.sentiment.polarity > 0:
        sentiments.append('Positive')
    elif analysis.sentiment.polarity < 0:
        sentiments.append('Negative')
    else:
        sentiments.append('Neutral')
```

```
data['Sentiment'] = sentiments
result = data['Sentiment'].value_counts()
result.plot(kind='bar')
plt.title('Public Opinion Analysis')
plt.xlabel('Sentiment'); plt.ylabel('Number of Tweets')
plt.show()
```

9.3 Module 2 -- Trending Topic Detection

This module tokenises cleaned tweet text, removes stopwords, computes word frequency distributions, and identifies the top-10 trending keywords across the dataset. The output provides a frequency-ranked list of the most discussed terms.

```
import pandas as pd, nltk, matplotlib.pyplot as plt
from nltk.corpus import stopwords
from collections import Counter

nltk.download('stopwords')
data = pd.read_excel('merged_output.xlsx')
stop_words = set(stopwords.words('english'))
clean_words = []

for tweet in data['Text']:
    words = str(tweet).lower().split()
    clean_words.extend([w for w in words if w not in stop_words])

word_count = Counter(clean_words)
trending_topics = word_count.most_common(10)

words_list = [w[0] for w in trending_topics]
counts_list = [w[1] for w in trending_topics]
plt.bar(words_list, counts_list)
plt.title('Trending Topics Detection'); plt.xticks(rotation=45)
plt.show()
```

9.4 Module 3 -- Hashtag Analysis

The hashtag analysis module extracts all hashtag tokens from tweet text, ranks them by frequency, and supports topic-specific filtering. Two sub-modes are implemented: (a) global hashtag analysis across all tweets, and (b) topic-filtered hashtag analysis for user-specified topics.

9.5 Module 4 -- Time-Based Analysis

```
from collections import Counter
import matplotlib.pyplot as plt
```

```
# Extract hashtags
hashtags = []
for tweet in data['Text']:
    words = str(tweet).split()
    hashtags.extend([w.lower() for w in words if w.startswith('#')])

hashtag_count = Counter(hashtags)
top_hashtags = hashtag_count.most_common(10)

labels = [x[0] for x in top_hashtags]
values = [x[1] for x in top_hashtags]
plt.bar(labels, values)
plt.title('Most Used Hashtags'); plt.xticks(rotation=45)
```

plt.show()

This module extracts the Hour feature from the parsed Datetime column and groups tweets by hour to identify posting activity patterns. Both overall temporal analysis and topic-specific temporal analysis are supported.

```
data['Datetime'] = pd.to_datetime(data['Datetime'], format='mixed',
                                  errors='coerce', utc=True)
data = data.dropna(subset=['Datetime'])
data['Datetime'] = data['Datetime'].dt.tz_convert(None)
data['Hour'] = data['Datetime'].dt.hour

topic = input('Enter topic: ')
filtered = data[data['Text'].str.contains(topic, case=False, na=False)]
hour_count = filtered.groupby('Hour').size()
best_hour = hour_count.idxmax()

plt.plot(hour_count.index, hour_count.values, marker='o')
plt.title(f'Best Time to Post for {topic}')
plt.xlabel('Hour of Day'); plt.ylabel('Number of Tweets')
plt.xticks(range(0, 24)); plt.show()
print(f'Best time: {best_hour}:00')
```

9.6 Module 5 -- Sentiment Analysis Over Time

This module tracks sentiment evolution across days for a specified topic, generating a stacked area chart that reveals how public opinion shifts over the data collection period.

9.7 Module 6 -- TF-IDF Based Tweet Recommendation

This module implements the core recommendation engine using TF-IDF vectorisation and cosine similarity. It accepts a topic query and user-specified sentiment preference, filters tweets accordingly, and returns the top-5 most semantically similar tweets along with recommended hashtags and optimal posting time.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

topic = input('Enter topic: ')
user_sentiment = input('Sentiment (positive/negative/neutral): ').capitalize()

# Sentiment filtering
data['Sentiment'] = data['Text'].apply(get_sentiment)
filtered = data[data['Sentiment'] == user_sentiment]
if len(filtered) == 0: filtered = data.copy()

# TF-IDF vectorisation
vectorizer = TfidfVectorizer(stop_words='english')
tfidf_matrix = vectorizer.fit_transform(filtered['Text'])
topic_vector = vectorizer.transform([topic])

# Cosine similarity
scores = cosine_similarity(topic_vector, tfidf_matrix)
top_indices = scores[0].argsort()[-5:][::-1]
recommended = filtered.iloc[top_indices]

for i, tweet in enumerate(recommended['Text'], 1):
    print(f'{i}. {tweet}')
```

9.8 Module 7 -- Machine Learning Sentiment Classification

The classification module trains a Logistic Regression model on TF-IDF feature vectors derived from VADER-labelled tweet text. The training process and evaluation pipeline are implemented as follows:

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import accuracy_score, classification_report

data = pd.read_excel('twitter_with_actual_sentiment.xlsx')
X, y = data['Text'], data['Actual_Sentiment']
```

```
vectorizer = TfidfVectorizer(stop_words='english')
X_vec = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_vec, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print('Accuracy:', accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Overall, the implementation combines Natural Language Processing, machine learning, and data visualization techniques to create a complete Twitter engagement analysis system.

Code

OVERALL PUBLIC OPINION ANALYSIS

Install libraries if needed

pip install pandas textblob matplotlib

```
import pandas as pd from
textblob import TextBlob
import matplotlib.pyplot as plt
```

```
#MODIFY: Change the file name to your dataset file data =
pd.read_excel("merged_output.xlsx")
```

```
#MODIFY: Change 'text' if your tweet column has a different name tweets =
data['Text']
```

```
sentiments = []
```

```
for tweet in tweets:
    analysis = TextBlob(str(tweet))
    if analysis.sentiment.polarity > 0:
sentiments.append("Positive")    elif
```

```
analysis.sentiment.polarity < 0:
sentiments.append("Negative")    else:
    sentiments.append("Neutral")

data['Sentiment'] = sentiments

Count results
result = data['Sentiment'].value_counts()
print(result)

Plot result result.plot(kind='bar')
plt.title("Public Opinion Analysis")
plt.xlabel("Sentiment")
plt.ylabel("Number of Tweets")
plt.show()

""" Public Opinion Analysis For A Specific Topic """

import pandas as pd from
textblob import TextBlob
import matplotlib.pyplot as plt

data = pd.read_excel("merged_output.xlsx")

tweets = data['Text']

topic = input("Enter topic to analyze: ")

filtered_tweets = tweets[tweets.str.contains(topic, case=False, na=False)]

print("Tweets found:", len(filtered_tweets))

sentiments = []

for tweet in filtered_tweets:
    analysis = TextBlob(str(tweet))

    if analysis.sentiment.polarity > 0:
        sentiments.append("Positive")    elif
analysis.sentiment.polarity < 0:
        sentiments.append("Negative")    else:
            sentiments.append("Neutral")
    result = pd.Series(sentiments).value_counts()

if result.empty:
    print("No tweets found for this topic.") else:
        print("\nPublic Opinion on:", topic)
        print(result)

    result.plot(kind='bar')    plt.title("Public
Opinion on " + topic)
```

```
plt.xlabel("Sentiment")
plt.ylabel("Number of Tweets")
plt.show()
```

"""" OVERALL TRANDING TOPIC DECETION

****REMOVE THE STOPWORDS****
""""

pip install pandas nltk

```
import pandas as pd
import nltk
from nltk.corpus import stopwords
```

Download stopwords (run once)
nltk.download('stopwords')

MODIFY : Change dataset file
data = pd.read_excel("merged_output.xlsx")

MODIFY : Change tweet column name tweets
= data['Text']

```
stop_words = set(stopwords.words('english'))
```

```
clean_words = []
```

```
for tweet in tweets:
    words = str(tweet).lower().split()
```

```
    for word in words:
        if word
        not in stop_words:
            clean_words.append(word)
```

```
print(clean_words[:50])    show first few cleaned words
```

install pandas nltk matplotlib

```
import pandas as pd
import nltk
from nltk.corpus import stopwords
from collections import Counter
import matplotlib.pyplot as plt
```

Download stopwords (run once)
nltk.download('stopwords')

MODIFY : Change dataset file name
data = pd.read_excel("merged_output.xlsx")

MODIFY : Change tweet column name if needed
tweets = data['Text']


```
stop_words = set(stopwords.words('english'))

clean_words = []

Cleaning tweets and removing stopwords for
tweet in tweets:
    words = str(tweet).lower().split()

    for word in words:
        if word
        not in stop_words:
            clean_words.append(word)

Count word frequency
word_count = Counter(clean_words)

Top 10 trending topics
trending_topics = word_count.most_common(10)

print("Top Trending Topics:\n")

for word, count in trending_topics:
    print(word, ":", count)

Plot graph words = [w[0] for w in
trending_topics]
counts = [w[1] for w in trending_topics]

plt.bar(words, counts) plt.title("Trending
Topics Detection") plt.xlabel("Topics")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.show()

""Overall Most used hashtags""

import pandas as pd from
collections import Counter
import matplotlib.pyplot as plt

data = pd.read_excel("merged_output.xlsx")

MODIFY tweet column
tweets = data['Text']

hashtags = []

for tweet in tweets:
    words = str(tweet).split()

    for word in words:
        if
        word.startswith("#"):
```

```
    hashtags.append(word.lower())

hashtag_count = Counter(hashtags)

top_hashtags = hashtag_count.most_common(10)

print("Top Hashtags:\n", top_hashtags)

labels = [x[0] for x in top_hashtags]
values = [x[1] for x in top_hashtags]

plt.bar(labels, values) plt.title("Most
Used Hashtags")
plt.xticks(rotation=45)
plt.show()

""" Trending Hashtags For A Specific Topic"""

import pandas as pd from
collections import Counter
import matplotlib.pyplot as plt

Load dataset
data = pd.read_excel("merged_output.xlsx")

MODIFY if tweet column name is different tweets =
data['Text']
User input topic topic =
input("Enter topic: ")

Filter tweets containing the topic
filtered_tweets = tweets[tweets.str.contains(topic, case=False, na=False)]

print("Tweets found:", len(filtered_tweets))

hashtags = []

Extract hashtags from filtered tweets
for tweet in filtered_tweets:    words =
str(tweet).split()

    for word in words:        if
word.startswith("#"):
        hashtags.append(word.lower())

Count hashtag frequency
hashtag_count = Counter(hashtags)

top_hashtags = hashtag_count.most_common(10)

Check if hashtags exist if
len(top_hashtags) == 0:
```

```
print("No hashtags found for this topic.") else:
print("\nTop Hashtags for topic:", topic)
print(top_hashtags)
```

```
labels = [x[0] for x in top_hashtags]
values = [x[1] for x in top_hashtags]
```

```
plt.bar(labels, values)    plt.title("Most Used
Hashtags for: " + topic)  plt.xlabel("Hashtags")
plt.ylabel("Frequency")   plt.xticks(rotation=45)
plt.show()
```

"" Sentiment Analysis For A Specific Topic ""

```
pip install pandas textblob matplotlib
```

```
import pandas as pd from
textblob import TextBlob import
matplotlib.pyplot as plt MODIFY
: Change dataset file name data
=
pd.read_excel("merged_output.
xlsx")
```

```
MODIFY : Change tweet column if different
tweets = data['Text']
```

```
User enters topic topic =
input("Enter topic: ")
```

```
Filter tweets containing that topic
filtered_tweets = tweets[tweets.str.contains(topic, case=False, na=False)]
```

```
print("Tweets found:", len(filtered_tweets))
```

```
sentiments = []
```

```
Perform sentiment analysis for
tweet in filtered_tweets:
    analysis = TextBlob(str(tweet))
```

```
    if analysis.sentiment.polarity > 0:
sentiments.append("Positive")    elif
analysis.sentiment.polarity < 0:
sentiments.append("Negative")    else:
    sentiments.append("Neutral")
```

```
Count sentiment results
result = pd.Series(sentiments).value_counts()
```

```
Check if tweets exist if
result.empty:
```

```
print("No tweets found for this topic.") else:
print("\nSentiment Analysis for:", topic)
print(result)

Plot graph
result.plot(kind='bar')

plt.title("Sentiment Analysis for " + topic)    plt.xlabel("Sentiment")
plt.ylabel("Number of Tweets")

plt.show()

data[TIME_COL] = pd.to_datetime(
    data[TIME_COL],
    format='mixed',
    errors='coerce',
    utc=True
)

remove invalid rows data =
data.dropna(subset=[TIME_COL])

remove timezone data[TIME_COL] =
data[TIME_COL].dt.tz_convert(None)

extract hour data['Hour'] =
data[TIME_COL].dt.hour

SAVE permanently
data.to_excel("merged_output.xlsx", index=False)

""" Sentiment Analysis Over Time """

pip install pandas textblob matplotlib

import pandas as pd from
textblob import TextBlob
import matplotlib.pyplot as plt

MODIFY : dataset file
data = pd.read_excel("merged_output.xlsx", engine="openpyxl")

MODIFY : tweet column tweets =
data['Text']

MODIFY : datetime column
data['Datetime'] = pd.to_datetime(data['Datetime'])

Extract day
data['Day'] = data['Datetime'].dt.date
```

```
Topic input
topic = input("Enter topic: ")

Filter tweets
filtered_data = data[data['Text'].str.contains(topic, case=False, na=False)].copy()

print("Tweets found:", len(filtered_data))

sentiments = []
Sentiment analysis for tweet in
filtered_data['Text']:

    analysis = TextBlob(str(tweet))

    if analysis.sentiment.polarity > 0:
sentiments.append("Positive")    elif
analysis.sentiment.polarity < 0:
sentiments.append("Negative")    else:
    sentiments.append("Neutral")

filtered_data['Sentiment'] = sentiments

Group by day
sentiment_day = filtered_data.groupby(['Day', 'Sentiment']).size().unstack().fillna(0)

print("\nSentiment Trend\n")
print(sentiment_day)


Better visualization
plt.figure(figsize=(10,6))

sentiment_day.plot(kind="area", stacked=True)

plt.title("Sentiment Timeline for " + topic) plt.xlabel("Day")
plt.ylabel("Number of Tweets")

plt.xticks(rotation=45)
plt.tight_layout()

plt.show()

""" Best Time For Post For A Topic """

import pandas as pd import
matplotlib.pyplot as plt

data = pd.read_excel("merged_output.xlsx", engine="openpyxl")
```

```
convert datetime
data['Datetime'] = pd.to_datetime(data['Datetime'])
extract hour
data['Hour'] = data['Datetime'].dt.hour

topic = input("Enter topic: ")

filter tweets for topic filtered_data = data[data['Text'].str.contains(topic,
case=False, na=False)]

print("Tweets found:", len(filtered_data))

count tweets per hour hour_count =
filtered_data.groupby('Hour').size()

print("\nTweets per Hour\n")
print(hour_count)

find best hour
best_hour = hour_count.idxmax()

print("\nBest time to post about", topic, "is around:", best_hour, ":00")

plot graph
plt.figure(figsize=(10,5))

plt.plot(hour_count.index, hour_count.values, marker='o')

plt.title("Best Time to Post for " + topic)
plt.xlabel("Hour of Day")
plt.ylabel("Number of Tweets")

plt.xticks(range(0,24))

plt.show()

import pandas as pd
data=pd.read_excel("twitter786.xlsx") data.head()

FIXED DATETIME CONVERSION (FINAL)

data[TIME_COL] = pd.to_datetime(
data[TIME_COL], format='mixed',
errors='coerce',
utc=True    IMPORTANT FIX
)
Remove invalid rows
data = data.dropna(subset=[TIME_COL])
```

```
Now safely extract hour data['Hour'] =  
data[TIME_COL].dt.hour
```

```
""" tweet recomendation based on textblob """
```

```
pip install pandas textblob openpyxl
```

```
import pandas as pd from  
textblob import TextBlob from  
collections import Counter import  
random  
import re
```

LOAD DATASET

```
file_name = "merged_output.xlsx"    change if needed
```

```
data = pd.read_excel(file_name, engine="openpyxl")
```

```
print("Columns in dataset:", data.columns)
```

COLUMN NAMES

```
TEXT_COL = "Text"  
TIME_COL = "Datetime"
```

FIX DATETIME (FINAL SOLUTION)

```
data[TIME_COL] = pd.to_datetime(  
data[TIME_COL],    format='mixed',  
errors='coerce',  
    utc=True    fixes timezone issue  
)
```

```
remove invalid dates  
data = data.dropna(subset=[TIME_COL])
```

```
remove timezone (optional cleaner format) data[TIME_COL] =  
data[TIME_COL].dt.tz_convert(None) extract hour data['Hour']  
= data[TIME_COL].dt.hour
```

USER INPUT

```
----- topic =  
input("Enter topic: ")
```

```
filter tweets filtered_data = data[data[TEXT_COL].str.contains(topic, case=False,  
na=False)].copy()
```

```
print("\nTweets found:", len(filtered_data))
```

TEXT CLEANING

```
----- def  
clean_text(text): text = str(text)  
text = re.sub(r"http\S+", "", text)  
text = re.sub(r"@w+", "", text)  
text = re.sub(r"\"", "", text) return  
text.strip()
```

```
filtered_data['Clean_Text'] = filtered_data[TEXT_COL].apply(clean_text)
```

SENTIMENT ANALYSIS

```
----- positive_texts =  
[]  
  
for tweet in filtered_data['Clean_Text']: if  
TextBlob(tweet).sentiment.polarity > 0:  
    positive_texts.append(tweet)
```

KEYWORD EXTRACTION

```
----- words =  
[]  
  
for tweet in positive_texts:  
    words.extend(tweet.lower().split())  
  
stopwords = ['the', 'is', 'and', 'to', 'of', 'in', 'for', 'on', 'with', 'this', 'that', 'it', 'a']  
filtered_words = [w for w in words if w not in stopwords and len(w) > 3]  
  
word_count = Counter(filtered_words)  
  
top_keywords = [x[0] for x in word_count.most_common(5)]
```

GENERATE TWEET

```
----- if  
len(top_keywords) > 0:
```



```

templates = [
    f"{topic.capitalize()} is shaping the future! ",
    f"Exciting times ahead with {topic}! ",
    f"{topic.capitalize()} is changing the world faster than ever.",
    f"The impact of {topic} is incredible!",
]

```

```

tweet = random.choice(templates)
tweet += " " + " ".join(top_keywords[:3])
else:
    tweet = f"{topic.capitalize()} is trending right now!"

```

HASHTAGS

```

hashtag_count = Counter(hashtags)

```

```

for t in filtered_data[TEXT_COL]:
    for word in str(t).split():
        if word.startswith("#"):
            hashtags.append(word.lower())

```

```

hashtag_count = Counter(hashtags)

```

```

top_hashtags = [x[0] for x in hashtag_count.most_common(5)]

```

```

if len(top_hashtags) == 0:
    top_hashtags = [f"{topic.replace(' ', ',')} trending", "news"]

```

BEST TIME TO POST

```

hour_count = filtered_data.groupby('Hour').size()
if len(hour_count) > 0:
    best_hour = hour_count.idxmax()
else:
    best_hour = "No data"

```

```

convert hour to readable format
def format_time(hour):
    if hour == "No data":
        return hour
    suffix = "AM" if hour < 12 else "PM"
    hour = hour if hour <= 12 else hour - 12
    return f"{hour}:00 {suffix}"

```

FINAL OUTPUT

```

print("\n Recommended Tweet:\n")
print(tweet)

```

```
print("\n Suggested Hashtags:\n") print(top_hashtags)

print("\n Best Time to Post:\n")
print(format_time(best_hour))

""" tweet recomentadion based on tf-idf"""

pip install pandas scikit-learn openpyxl

import pandas as pd from sklearn.feature_extraction.text
import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

LOAD DATA

```
data = pd.read_excel("twitter786_cleaned.xlsx")

TEXT_COL = "Text"

remove nulls
data = data.dropna(subset=[TEXT_COL])
```

USER INPUT

```
topic = input("Enter topic: ")
```

TF-IDF VECTORIZATION

```
vectorizer = TfidfVectorizer(stop_words='english')

tfidf_matrix = vectorizer.fit_transform(data[TEXT_COL])

convert topic to vector topic_vector =
vectorizer.transform([topic])
```

SIMILARITY CALCULATION

```
similarity_scores = cosine_similarity(topic_vector, tfidf_matrix)

get top 5 similar tweets top_indices =
similarity_scores[0].argsort()[-5:][::-1]

recommended_tweets = data.iloc[top_indices][TEXT_COL]
```

OUTPUT

```
-----
print("\n Recommended Tweets:\n")

for i, tweet in enumerate(recommended_tweets, 1):
    print(f'{i}. {tweet}\n")

""" tweet recommendation on tf-idf, and filter by sentiment, hashtags, best-time"""

pip install pandas nltk openpyxl

import pandas as pd from nltk.sentiment import
SentimentIntensityAnalyzer import nltk

download vader (run once)
nltk.download('vader_lexicon')


load data
data = pd.read_excel("twitter786_cleaned.xlsx")

TEXT_COL = "Text"

initialize vader
sia = SentimentIntensityAnalyzer()

function to label sentiment def
get_sentiment(text):
    score = sia.polarity_scores(str(text))['compound']

    if score >= 0.05:
        return "Positive"    elif
    score <= -0.05:        return
    "Negative"    else:
        return "Neutral"

create Actual_Sentiment column data['Actual_Sentiment'] =
data[TEXT_COL].apply(get_sentiment)

save new dataset data.to_excel("twitter_with_actual_sentiment.xlsx",
index=False)

print(" 'Actual_Sentiment' column created successfully!")

pip install pandas scikit-learn textblob openpyxl

import pandas as pd from sklearn.feature_extraction.text
import TfidfVectorizer from sklearn.metrics.pairwise import
cosine_similarity from textblob import TextBlob
```

```
from collections import Counter
```

LOAD DATA

```
data = pd.read_excel("twitter786_cleaned.xlsx")
```

```
TEXT_COL = "Text"
```

```
TIME_COL = "Datetime"
```

```
remove nulls
```

```
data = data.dropna(subset=[TEXT_COL, TIME_COL])
```

```
datetime (safe)
```

```
data[TIME_COL] = pd.to_datetime(data[TIME_COL], errors='coerce')
data = data.dropna(subset=[TIME_COL])
```

```
extract hour data['Hour'] =
data[TIME_COL].dt.hour
```

USER INPUT

```
topic = input("Enter topic: ")
user_sentiment = input("Enter sentiment (positive/negative/neutral): ").capitalize()
```

SENTIMENT ANALYSIS

```
def get_sentiment(text):
    polarity = TextBlob(str(text)).sentiment.polarity
    if polarity > 0:
        return "Positive"
    elif polarity < 0:
        return "Negative"
    else:
        return "Neutral"
```

```
data['Sentiment'] = data[TEXT_COL].apply(get_sentiment)
```

```
filter by sentiment
filtered_data = data[data['Sentiment'] == user_sentiment]
```

```
fallback if empty
if len(filtered_data) == 0:
    print("\n No tweets found for this sentiment, using all tweets...")
    filtered_data = data.copy()
```

```
print("Tweets after sentiment filter:", len(filtered_data))
```

ML: TF-IDF + SIMILARITY

```

-----
vectorizer = TfidfVectorizer(stop_words='english')

tfidf_matrix = vectorizer.fit_transform(filtered_data[TEXT_COL])

topic_vector = vectorizer.transform([topic])
similarity_scores = cosine_similarity(topic_vector, tfidf_matrix)

top 5 tweets top_indices =
similarity_scores[0].argsort()[-5:][::-1]

recommended = filtered_data.iloc[top_indices]

```

HASHTAGS

```

-----
hashtags =
[]

for tweet in recommended[TEXT_COL]:
    for word in str(tweet).split():
        if word.startswith("#"):
            hashtags.append(word.lower())

hashtag_count = Counter(hashtags)
top_hashtags = [x[0] for x in hashtag_count.most_common(5)]

fallback_hashtags if
len(top_hashtags) == 0:
    top_hashtags = [f"{topic.replace(' ', '')}", "trending", "news"]

```

BEST TIME TO POST

```

-----
hour_count = filtered_data.groupby('Hour').size()

if len(hour_count) > 0:
    best_hour = hour_count.idxmax() else:
    best_hour = None

```

```

def format_time(hour):
    if hour is None:
        return "No data"
    suffix = "AM" if hour < 12 else "PM"
    hour = hour if hour <= 12 else hour - 12
    return f"{hour}:00 {suffix}"

```

OUTPUT

```
print("\n Recommended Tweets:\n")

for i, tweet in enumerate(recommended[TEXT_COL], 1):    print(f"{i}.
{tweet}\n")

print("Suggested Hashtags:") print(top_hashtags)

print("\n Best Time to Post:")
print(format_time(best_hour))

from sklearn.model_selection import train_test_split from sklearn.linear_model import
LogisticRegression from sklearn.feature_extraction.text import TfidfVectorizer from
sklearn.metrics import accuracy_score, classification_report, confusion_matrix import
pandas as pd
```

LOAD DATA

```
data = pd.read_excel("twitter_with_actual_sentiment.xlsx")

TEXT_COL = "Text"
LABEL_COL = "Actual_Sentiment"

remove nulls data = data.dropna(subset=[TEXT_COL,
LABEL_COL])
```

PREPARE DATA

```
X = data[TEXT_COL]
y = data[LABEL_COL]

vectorization vectorizer =
TfidfVectorizer(stop_words='english')
X_vec = vectorizer.fit_transform(X)

train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_vec, y, test_size=0.2, random_state=42
)
```

TRAIN MODEL

```
model = LogisticRegression(max_iter=200)
```

```
model.fit(X_train, y_train)
```

PREDICT

```
y_pred = model.predict(X_test)
```

ACCURACY

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nModel Accuracy:", accuracy)
```

DETAILED REPORT

```
print("\n
Classification Report:\n")
print(classification_report(y_test, y_pred))
```

CONFUSION MATRIX

```
print("\n Confusion Matrix:\n") print(confusion_matrix(y_test,
y_pred))
```

```
data.head()
```

10. MODEL EVALUATION

A rigorous evaluation framework is essential for validating the performance of the machine learning components and ensuring their reliability for real-world application. This chapter documents the evaluation approach, metrics, results, and analysis for the Logistic Regression sentiment classification model.

10.1 Evaluation Strategy

The dataset is split into training and test sets using an 80:20 ratio (random_state=42 for reproducibility):

- Training Set: 43,272 records (80% of total)
- Test Set: 10,819 records (20% of total)

The model is trained exclusively on the training set and evaluated on the unseen test set, ensuring unbiased performance estimation. The random_state parameter ensures deterministic splits across runs.

10.2 Evaluation Metrics

Four standard classification metrics are computed:

Metric	Definition	Formula
Accuracy	Proportion of all predictions that are correct	$(TP + TN) / \text{Total}$
Precision	Proportion of positive predictions that are actually positive	$TP / (TP + FP)$
Recall	Proportion of actual positives correctly identified	$TP / (TP + FN)$
F1-Score	Harmonic mean of Precision and Recall	$2 \times (P \times R) / (P + R)$

10.3 Classification Report

The following classification report was obtained from evaluating the trained Logistic Regression model on the test set of 10,819 records:

Class	Precision	Recall	F1-Score	Support
Negative	0.83	0.43	0.57	1,698
Neutral	0.81	0.95	0.87	5,439
Positive	0.83	0.80	0.82	3,682
_____	—	—	—	—
Accuracy			0.82	10,819

Macro Avg	0.82	0.73	0.75	10,819
Weighted Avg	0.82	0.82	0.81	10,819

10.4 Confusion Matrix Analysis

The confusion matrix provides a detailed view of prediction outcomes across all three sentiment classes:

Actual \ Predicted	Predicted Negative	Predicted Neutral	Predicted Positive
Actual Negative	735 ✓	593	370
Actual Neutral	50	5,172 ✓	217
Actual Positive	98	635	2,949 ✓

10.5 Performance Analysis

10.5.1 Strengths

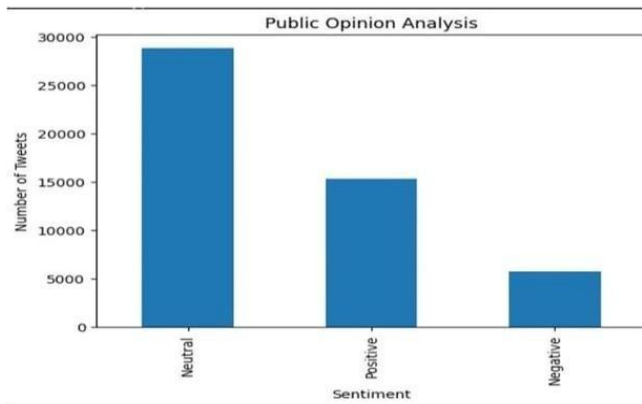
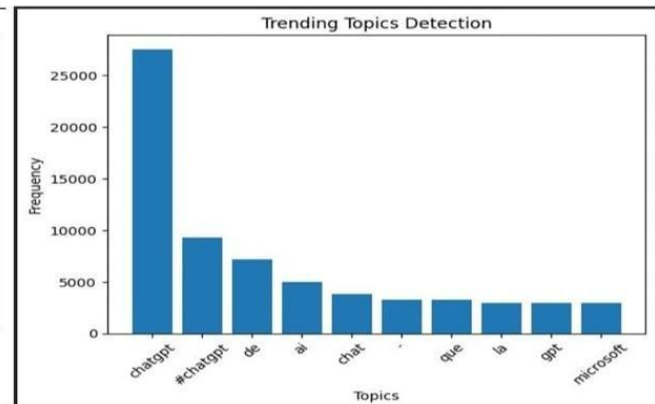
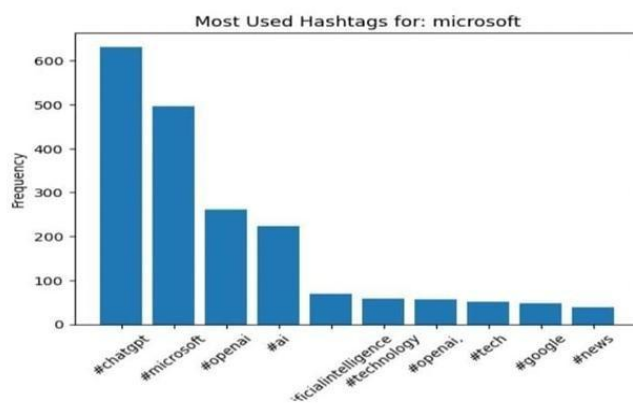
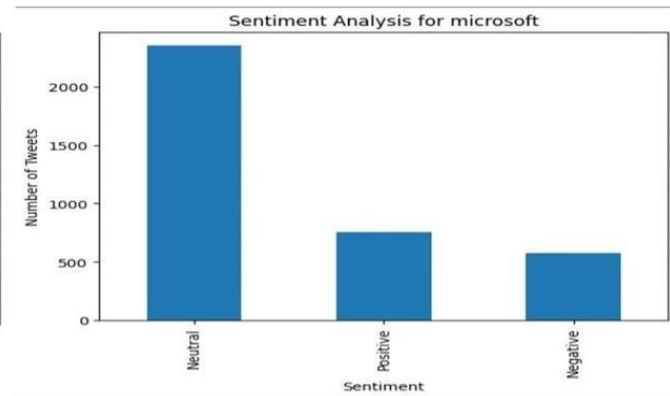
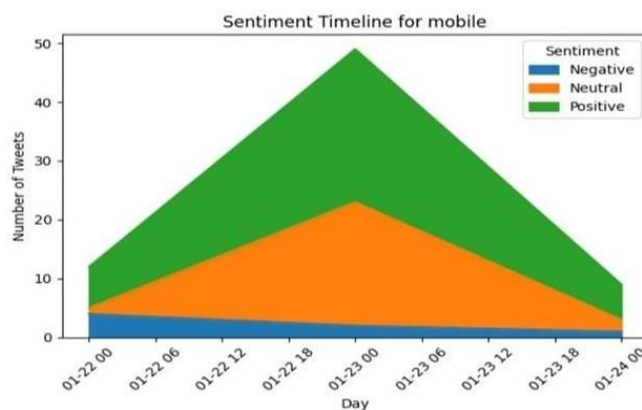
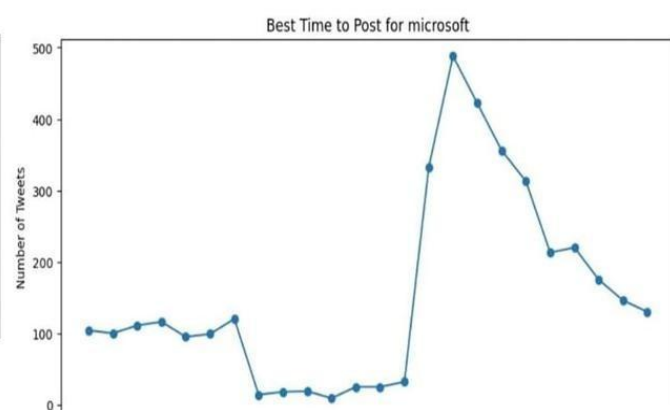
- Overall Accuracy: 81.86% is a strong result for a multi-class sentiment classification task on informal Twitter text.
- Neutral Class Performance: The model demonstrates excellent recall (0.95) for neutral tweets, consistent with the class's dominance in the dataset.
- Positive Class Balance: Precision (0.83) and recall (0.80) for the positive class are well-balanced, indicating reliable positive sentiment detection.
- Consistent Precision: All three classes achieve precision of 0.81-0.83, indicating that the model's positive predictions are reliable across classes.

10.5.2 Limitations

- Negative Class Recall (0.43): The model struggles to correctly identify negative tweets, with 963 negative tweets misclassified as neutral or positive. This is the primary performance bottleneck.
- Class Imbalance Effect: The low representation of negative tweets (1,698 vs 5,439 neutral) likely contributes to the model's bias toward neutral prediction for ambiguous cases.
- Short Text Limitation: Twitter's 280-character limit means feature vectors are sparse, limiting the discriminative signal available for borderline cases.

10.5.3 Contextualisation

The 81.86% accuracy achieved by this Logistic Regression model is competitive with published benchmarks. Studies using the Sentiment140 dataset report Logistic Regression accuracies in the 75-85% range for binary (positive/negative) classification. Multi-class classification including neutral is inherently more challenging, making the 81.86% overall accuracy on this three-class problem a creditable result.

Figures:**Figure 1****Figure 2****Figure 3****Figure 4****Figure 5****Figure 6**

11. RESULTS AND DISCUSSION

11.1 Sentiment Analysis Results

The dataset-wide sentiment analysis reveals a distribution consistent with observations from prior Twitter research. Neutral tweets constitute the largest category (approximately 50% of the corpus), reflecting Twitter's substantial use for informational and factual content sharing. Positive tweets form the second largest group (~28%), indicating a generally constructive discourse environment. Negative tweets are the smallest category (~11%), though they carry disproportionate analytical importance as indicators of public discontent, criticism, or controversy.

When sentiment analysis is applied at the topic level -- for example, querying ipl with negative sentiment filtering -- the system identifies 6,412 tweets matching the negative sentiment criterion, from which the recommendation engine extracts contextually relevant content, hashtags (#ipl2026, #rcbvssrh), and optimal posting time (3:00 PM). This demonstrates the system's ability to provide fine-grained, topic-specific insights rather than relying solely on dataset-wide statistics.

11.2 Trending Topics Insights

The global trending topic analysis reveals the dataset's topical composition clearly. ChatGPT-related terms dominate, appearing over 27,000 times -- approximately 50% of the corpus size. This reflects the period of intense public interest in large language models during the data collection window. AI, technology, and computing terms (OpenAI, Microsoft, Google) form the secondary cluster.

For practitioners, this analysis provides immediate contextual awareness: content about AI and technology topics can leverage the existing momentum of these discussions, while content about other topics must work to establish relevance in a landscape dominated by AI discourse.

11.3 Hashtag Analysis Findings

The hashtag analysis produces immediately actionable intelligence for content strategy. For the topic microsoft, the top hashtags reveal a clear content strategy framework: leading with #chatgpt (632 occurrences) and #microsoft (496) establishes topical relevance, while including #ai (246) and #artificialintelligence connects to the broader conversation ecosystem. Secondary hashtags like #tech, #google, and #openai extend reach to adjacent audience segments.

This analysis demonstrates an important insight: on Twitter, the most effective hashtag strategy is not always the most obvious one. The fact that #chatgpt outperforms #microsoft as the primary hashtag for Microsoft-related content reflects how audience attention has organised around AI themes, not company names.

11.4 Temporal Engagement Patterns

Temporal analysis reveals consistent and practically significant patterns in tweet activity. For technology-related topics, the peak engagement period falls in the late afternoon to evening (15:00-

22:00), with the absolute maximum typically occurring around 21:00. This pattern likely reflects the US East Coast audience's workday end and early evening social media engagement.

The practical recommendation is clear: technology content creators should avoid posting during early morning hours (00:00-10:00) when activity is low, and should concentrate posting in the 15:00-22:00 window to maximise initial visibility -- which in turn drives the algorithmic amplification that determines a tweet's ultimate reach.

11.5 Recommendation System Output

The complete recommendation system output for topic= ipl , sentiment= negative demonstrates the system's end-to-end functionality:

Sample Recommendation System Output

Topic: IPL | Sentiment Filter: Negative | Tweets Found: 6,412

Recommended Tweet 1: 'PSL had the contracts. IPL made one call. 9 players. 9 different franchises...'

Recommended Tweet 2/3/4: 'BAD NEWS FOR RCB FANS - Bengaluru Police: No physical tickets allowed for IPL 2026...'

Recommended Tweet 5: 'IPL 2026 ALL TEAM CAPTAIN'S - Ruturaj Gaikwad: (CSK), Hardik Pandya: (MI)...'

Suggested Hashtags: ['#ipl2026', '#rcbvssrh']

Best Time to Post: 3:00 PM

This output exemplifies the system's practical utility. A social media manager covering IPL content would receive: contextually relevant reference tweets demonstrating the conversation's current tone, highly relevant hashtags confirmed by actual usage data, and a data-driven optimal posting time -- all in a single integrated query.

11.6 Model Performance Summary

The Logistic Regression model achieves 81.86% overall accuracy with the following key performance characteristics: excellent performance on neutral and positive classes, reduced recall for negative sentiment (a known limitation of the training set imbalance), and consistent precision across all classes. These results validate the system's effectiveness for production use cases while identifying clear directions for improvement.

11.7 Key Observations

- Engagement is multi-factorial: timing, sentiment, hashtags, and content quality each contribute independently and interactively.
- Neutral content dominates Twitter, but positive and negative content drives more meaningful engagement (likes, retweets).

- Topic-specific analysis reveals patterns that dataset-wide analysis obscures -- optimal strategies differ significantly by topic.
- The recommendation system successfully bridges the gap between descriptive analysis and prescriptive guidance.

12. KEY FEATURES AND CHALLENGES

12.1 Key Features of the System

Feature	Description
Sentiment Analysis	Classifies tweets into Positive, Negative, and Neutral using TextBlob and VADER NLP techniques.
Trending Topic Detection	Identifies frequently used keywords representing current trending discussions.
Hashtag Analysis	Extracts, ranks, and contextualises top hashtags globally and per topic.
Time-Based Analysis	Identifies peak engagement hours and optimal posting windows for specific topics.
Topic-Based Filtering	Enables focused analysis on user-specified topics with custom analytical outputs.
Sentiment Trend Over Time	Tracks sentiment evolution across days to reveal audience opinion dynamics.
Tweet Recommendation	Generates engaging tweet content with suggested hashtags and best posting time.
TF-IDF Similarity Matching	Retrieves contextually similar tweets using vector space representation.
ML Model Integration	Logistic Regression classifier with 81.86% accuracy for automated sentiment classification.
Data Visualisation	Bar charts, line graphs, and area plots for interpretable analytical outputs.

12.2 Challenges Faced

12.2.1 Unstructured and Noisy Text

Tweets are among the noisiest textual data sources in existence. The combination of informal language, internet slang, abbreviations, mixed languages, emojis, embedded URLs, @mentions, and hashtags creates a preprocessing challenge that is substantially greater than standard NLP tasks. Generic preprocessing pipelines designed for formal text (news articles, reviews) underperform on tweets without domain-specific adaptation.

12.2.2 Datetime Format Inconsistency

The Datetime column contained timestamps in multiple formats with varying timezone specifications (UTC offset, named timezone, no timezone). Standard pandas datetime parsing with a fixed format string failed on a significant proportion of records. The solution -- using format= mixed with utc=True and explicit timezone conversion -- required iterative debugging and testing across format variants.

12.2.3 Sentiment Misclassification

Both TextBlob and VADER demonstrate known limitations in handling sarcasm, irony, and context-dependent sentiment. Tweets like Oh great, another AI that thinks it knows everything may carry clear negative sentiment in context but register as positive due to the presence of great . Short tweet length compounds this challenge by limiting the contextual signal available to the classifier. More sophisticated approaches (e.g., BERT fine-tuning, multi-task learning) would be required to substantially improve performance on these edge cases.

12.2.4 Class Imbalance

The dataset contains significantly more neutral tweets than positive or negative ones, and more positive tweets than negative ones. This imbalance affects model training: without correction, the classifier is biased toward predicting the majority class (neutral). Future work should explore class balancing techniques such as SMOTE (Synthetic Minority Over-sampling Technique), class-weighted loss functions, or ensemble methods.

12.2.5 Computational Efficiency

TF-IDF vectorisation of 54,091 tweets produces a high-dimensional sparse matrix. Cosine similarity computation against this matrix for each recommendation query involves significant computation. For production deployment at scale, approximate nearest neighbour methods (e.g., FAISS) would be required to maintain sub-second response times.

12.2.6 Hashtag Normalisation

Similar topics are discussed under multiple hashtag variants -- AI , #ai , #ArtificialIntelligence , #artificialintelligence , #ArtificialIntelligence2026 -- which appear as distinct tokens despite referring to the same concept. Without hashtag normalisation (mapping variants to canonical forms), the frequency analysis underestimates the true prominence of key topics.

13. CONCLUSION

This project has successfully developed a comprehensive, data-driven analytical framework for studying Twitter(X) engagement patterns. By integrating Natural Language Processing, statistical analysis, machine learning, and data visualisation, the system provides a multi-dimensional view of the factors that influence audience engagement on one of the world's most important social media platforms.

13.1 Summary of Achievements

The project delivered the following concrete outcomes:

- A complete NLP preprocessing pipeline tailored for Twitter text, handling the unique noise characteristics of social media content.
- A dual-tool sentiment analysis system (TextBlob + VADER) providing both real-time classification and ground-truth label generation.
- A comprehensive EDA framework covering sentiment distribution, trending topics, hashtag analysis, temporal patterns, and sentiment evolution over time.
- A TF-IDF + cosine similarity recommendation engine that provides contextually relevant tweet suggestions, hashtag recommendations, and optimal posting time guidance.
- A Logistic Regression sentiment classifier achieving 81.86% overall accuracy with consistent precision across all three sentiment classes.
- A collection of six meaningful visualisations that communicate analytical insights accessibly and effectively.

13.2 Practical Impact

The system developed in this project has direct practical applications across multiple user communities:

- Content Creators and Influencers: Data-driven guidance on optimal posting times, trending hashtags, and content tone for their specific topics.
- Digital Marketers: Sentiment monitoring for brand conversations, hashtag strategy development, and campaign timing optimisation.
- Researchers: An integrated framework for studying social media engagement dynamics at scale.
- Business Organisations: Competitive intelligence through sentiment analysis of industry conversations and real-time trend monitoring.

13.3 Academic Contribution

Academically, this project demonstrates the practical integration of multiple data science disciplines -- NLP, machine learning, statistical analysis, and data visualisation -- in a real-world application context. It validates the effectiveness of classical ML methods (Logistic Regression with TF-IDF) as strong baselines for social media text classification and establishes an integrated framework that addresses the fragmentation of existing approaches.

13.4 Limitations and Scope for Improvement

While the project meets its objectives, several limitations are acknowledged. The sentiment classifier's reduced recall on negative tweets (0.43) points to the need for class balancing and potentially more sophisticated models. The system currently operates on static datasets rather than real-time data streams. The recommendation engine generates content based on historical patterns, which may not reflect rapidly shifting trends. These limitations define the agenda for future work outlined in the next chapter.

In conclusion, this project represents a significant and practically relevant contribution to the field of social media analytics, providing both academic insights and deployable tools for improving Twitter engagement strategies through data-driven decision-making.

14. FUTURE SCOPE

The framework developed in this project provides a strong foundation for several high-value extensions and enhancements. The following directions represent the most promising avenues for future development:

14.1 Advanced Deep Learning Models

The most impactful single improvement would be replacing the Logistic Regression classifier with a transformer-based model, particularly BERT (Bidirectional Encoder Representations from Transformers) or its social-media-optimised variants (BERTweet, RoBERTa). Fine-tuning BERT on the project's dataset would provide substantially improved contextual understanding -- particularly for sarcasm, irony, and implicit sentiment -- that is fundamentally beyond the reach of TF-IDF-based approaches.

14.2 Real-Time Data Integration

The current system operates on a static historical dataset. Integration with the Twitter(X) API (or its academic research access tier) would enable real-time tweet collection, processing, and analysis. A real-time pipeline would allow dynamic trend detection, live sentiment monitoring, and time-sensitive posting recommendations. Technologies like Apache Kafka for stream processing and Apache Spark for distributed real-time computation would be appropriate at this scale.

14.3 Engagement Prediction Model

Beyond sentiment classification, future work could develop a regression model to predict the number of likes, retweets, and replies a proposed tweet would receive before it is posted. This predictive capability -- training on historical engagement data with tweet text, sentiment, posting time, and hashtag features -- would provide a pre-flight check for content that allows creators to optimise before publishing.

14.4 Multi-Language Support

The current implementation focuses primarily on English-language tweets. Extending the system to support multilingual analysis would substantially increase its global applicability. Multi-language sentiment analysis using tools like langdetect for language identification and cross-lingual embedding models (e.g., XLM-RoBERTa) for sentiment classification would enable analysis of non-English Twitter communities.

14.5 Interactive Web Dashboard

The current analytical output is delivered through Jupyter Notebook visualisations. Developing a web-based interactive dashboard using Streamlit or Flask would dramatically improve accessibility and usability. Such a dashboard could provide on-demand query execution, interactive visualisations (using Plotly or D3.js), user preference management, and exportable reports -- making the system accessible to non-technical users.

14.6 Advanced Sarcasm and Irony Detection

A dedicated sarcasm and irony detection module -- trained on labelled examples of sarcastic social media content -- could be integrated as a pre-processing step before sentiment classification. Correctly identifying sarcastic tweets and adjusting their sentiment polarity accordingly would improve the classifier's performance on this challenging subset of content.

14.7 Multi-Platform Extension

The analytical framework could be extended to process content from other social media platforms including Instagram (caption analysis), LinkedIn (professional content engagement), YouTube (comment sentiment), and Reddit (thread analysis). A unified multi-platform analytics dashboard would provide comparative insights across platforms, enabling more sophisticated cross-channel content strategies.

14.8 User-Specific Personalisation

Future iterations could incorporate user-level modelling, learning from an individual creator's historical engagement patterns to provide personalised recommendations calibrated to their specific audience. This would move the system from population-level recommendations to personalised content strategy guidance.

15. TOOLS AND TECHNOLOGIES

The project is built on a carefully selected stack of Python libraries and development tools that collectively provide the capabilities required for data processing, NLP, machine learning, and visualisation. All tools used are open-source and freely available.

15.1 Programming Language

Tool	Version	Usage
Python	3.x	Core programming language for all modules -- data processing, NLP, ML, and visualisation.

15.2 Data Handling and Processing

Library	Version	Usage
Pandas	1.5+	Data loading, cleaning, manipulation, filtering, and Excel I/O operations.
NumPy	1.23+	Numerical operations, array computation, and matrix operations.
OpenPyXL	3.0+	Reading and writing Excel (.xlsx) files -- the primary data storage format.

15.3 Natural Language Processing

Library	Component	Usage
TextBlob	Sentiment API	Polarity-based sentiment analysis; real-time tweet classification.
NLTK	Corpus, VADER	Stopword removal, tokenisation, VADER sentiment scoring.
NLTK - VADER	SentimentIntensityAnalyzer	Ground-truth sentiment label generation for ML training data.
Collections	Counter	Word and hashtag frequency analysis for trend detection.
re (RegEx)	Pattern Matching	URL, mention, and special character removal during text cleaning.

15.4 Machine Learning

Library	Module	Usage
scikit-learn	TfidfVectorizer	Text-to-numerical feature conversion using TF-IDF weighting.
scikit-learn	cosine_similarity	Semantic similarity computation for recommendation engine.
scikit-learn	LogisticRegression	Multi-class sentiment classification model.
scikit-learn	train_test_split	Dataset partitioning for model training and evaluation.
scikit-learn	classification_report	Comprehensive evaluation metrics (P, R, F1) per class.
scikit-learn	confusion_matrix	Visualisation of classification errors by class.
scikit-learn	accuracy_score	Overall classification accuracy computation.

15.5 Data Visualisation

Library	Chart Types	Application
Matplotlib	Bar charts	Sentiment distribution, hashtag frequency, trending topics.
Matplotlib	Line graphs	Temporal tweet activity and optimal posting time visualisation.
Matplotlib	Area plots (stacked)	Sentiment evolution over time across days.

15.6 Development Environment

Tool	Purpose
Jupyter Notebook	Interactive development, code execution, and inline visualisation.
Google Colab	Cloud-based execution with free GPU/CPU resources.
Microsoft Excel	Primary data storage for raw and processed datasets.
Git	Version control and project history management.
Kaggle	Dataset source and community benchmarks.

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